
DEVELOPMENT OF SOFTWARE APPLICATIONS I
COM-16203

Credits

8.0 (4 hrs lecture)

Lead Instructor

Alejandra Flores Mosri

Textbook

Notas de desarrollo de aplicaciones informáticas, F. López, ITAM

Supplemental materials

C# for Java Developers. Allen Jones, Adam Freeman. Microsoft Press. ISBN: 9780735617797.

Microsoft Visual Studio 2012 Unleashed. Mike Snell, Lars Powers. Sams Publishing. ISBN: 978-0672336256.

Media Beginning Windows Phone App Development. Henry Lee, Eugene Chuvyrov. Apress. ISBN-10: 1430241349.

Course Description

This course aims to teach students the ability to solve problems in a methodical and organized fashion. The students will develop information systems using current information technologies and attending to realistic business requirements.

Prerequisites

Data structures (COM 11102)

Major Status

Required

Course Outcomes

1. Establish a practical link between information technology and business requirements in an organization through system development.
2. Achieve the integration of MS Excel and Dynamic Link Libraries created in C# so as to perform complex data analysis.
3. Model, design and create databases so that they can be used in information systems.

4. Design and develop web-based information systems using ASPs and C#.
5. Design and develop mobile applications using Windows Phone 7.
6. Describe the various cloud computing models.

Student Outcomes

- h1) Awareness of current trends in the engineering discipline.
- h2) Awareness of the historical impact of engineering solutions.
- h3) Understanding of the impact of engineering solutions in a global context
- h4) Understanding of the impact of engineering solutions in an economic context
- h5) Understanding of the impact of engineering solutions in an environmental context
- h6) Understanding of the impact of engineering solutions in a societal context

Course Topics

- Client server and system architectures concepts.
- Database modeling, design, creation and management using SQL.
- Web-based system development using ASPs and C#.
- MS Excel and Dynamic Link Libraries integration using C#.
- Mobile application development using Windows Phone 7 and C#.